



SamanaHook

Impermanent loss insurance for committed liquidity

THE PROBLEM

IL is worst when depth is needed most

Token launch

Depeg

Post-exploit

Token unlock

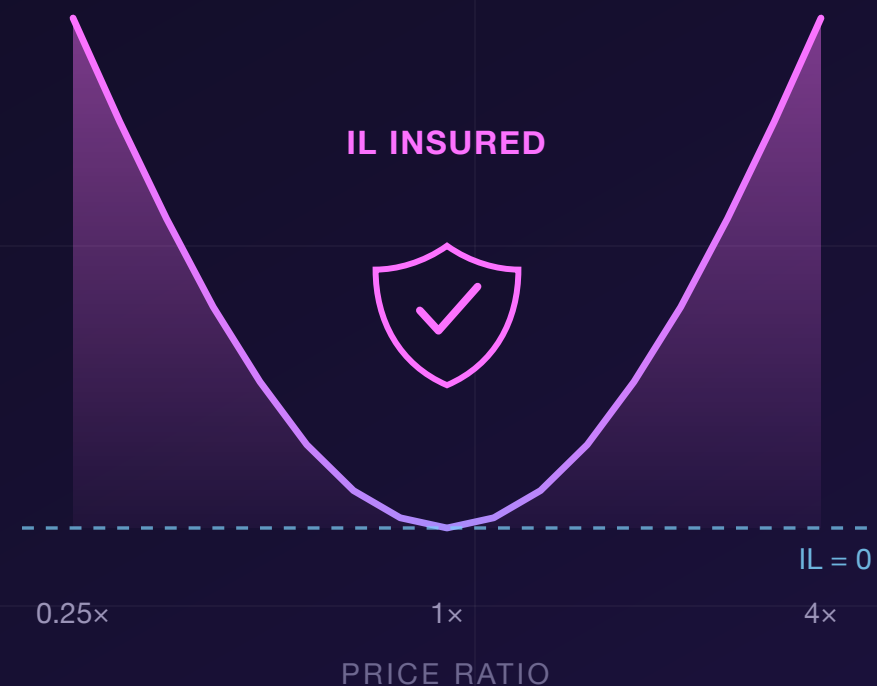
- Volatility is at its peak, so expected IL is at its peak
- Fee revenue is at its thinnest, nothing offsets the IL
- So rational LPs walk away, just when every unit of depth counts double

Staying in the pool is irrational. Someone has to pay LPs to stay.

THE FIX

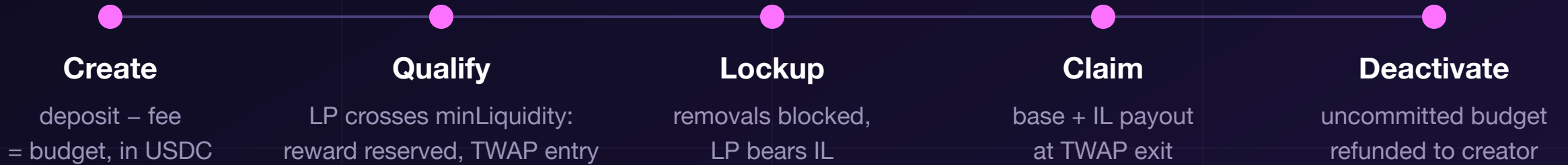
IL becomes a payout, not a loss

$$\text{payout} = \text{reward} \times \text{IL} \times \text{coverage}$$



HOW IT WORKS

The hook enforces the deal end to end



- No external oracle: entry and exit prices are the pool's own 30-min TWAP
- Pull-payments: claims survive deactivation, no reentrancy in callbacks
- Creator cost is bounded: payouts capped by the budget, worst case known upfront

Whoever needs the depth funds the bounty

New pool cold-start

PROTOCOL / DAO

Lockup: 7-30 days

Coverage: 50-100%

Budget: 2× rewards

Depeg defence

ISSUER TREASURY

Lockup: 3-14 days

Coverage: 100%

Budget: 3× rewards

Token unlock

FOUNDATION / WHALE

Lockup: 1-7 days

Coverage: 0-50%

Budget: 1.5× rewards

Same hook, same formula - only the knobs change

Real transactions, three acts

ACT 1

Trust the lockup

- Create 15 USDC bounty
- LP qualifies, 5 USDC reserved
- Early claim reverts on-chain:
`LiquidityStillLocked`
- Claim succeeds after lockup

ACT 2

Budget rules

- Second LP drains the budget
- Third LP qualifies but gets nothing: exhaustion is silent
- Deactivate refunds the uncommitted 3.5 USDC

ACT 3

IL insurance pays

- New bounty, 100% coverage
- Full-range LP qualifies
- Big swap moves the price, creating real IL
- Claim pays base + IL payout, more than the 5 USDC base

SamanaHook @ `0x597cb94f36f8ECA3a450c7f13C237e4D667E9680` · Sepolia

Committed liquidity, insured



Tested end to end: 100% line coverage, Sepolia fork suite, live deployment

WHAT'S NEXT

Pro-rata collective bounties · ERC1155 position tokens

Real LP attribution via hookData

github.com/diego-G/samana-hook